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Name _____

Nov. 7, 2003 Sunday

Department _____

SID _____

University Physics
Midterm Examination

School of Intensive Instruction in Science and Arts, Nanjing University

Physical Constants

<i>Electron charge</i>	$e = 1.60 \times 10^{-19} \text{ C}$
<i>Dielectric constant of free space</i>	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
<i>Permeability of free space</i>	$\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$
<i>Vacuum speed of light</i>	$c = 3.00 \times 10^8 \text{ m/s}$
<i>Electron mass</i>	$m_e = 9.11 \times 10^{-31} \text{ kg}$
<i>Proton mass</i>	$m_p = 1.67 \times 10^{-27} \text{ kg}$
<i>Electron volt (eV)</i>	$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$

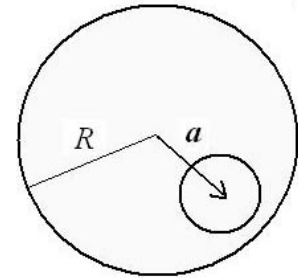
Select five out of following six problems.

1 (20pts). A nonconducting sphere of radius R carries a uniform charge per unit volume ρ . Let $\mathbf{r}$ be the vector from the center of the sphere to a point P .

a. Show that the electric field at P is given by $\mathbf{E} = \rho \mathbf{r} / 3\epsilon_0$ when the point P is within the sphere.

b. Show that the electric field at P is given by $\mathbf{E} = \rho R^3 \mathbf{r} / 3\epsilon_0 r^3$ when the point P is outside the sphere.

c. A spherical cavity is created in the sphere, as shown in the accompanying figure. In this case, show that the electric field at all points within the cavity is $\mathbf{E} = \rho \mathbf{a} / 3\epsilon_0$ (uniform field), where $\mathbf{a}$ is the vector connecting the center of the charged sphere and the center of the cavity.



2.(20pts). A uniformly charged rod of length L with linear charge density λ is placed on the z -axis from $z = -L/2$ to $z = L/2$.

(1) Obtain the electrical potential $V(x, y)$ for any point on the x - y plane.

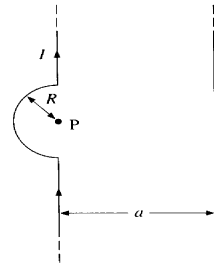
(2) What would you expect the potential to be for $\sqrt{x^2 + y^2} \gg L$?

(3) What is the electric field in the x - y plane in the limit that $\sqrt{x^2 + y^2} \ll L$?

3. (20pts). Two long wires, one of which has a semicircular bend of radius R , are positioned as shown in the accompanying figure. Both wires carry a current I .

(a) How far apart must they be so that the resultant magnetic field at the point P is zero?

(b) Does the current in the straight wire flow up or down?



4.(20pts). A circular wire loop of radius r falls under the influence of gravity and in the presence of a magnetic field given by

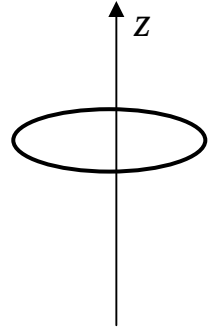
$$\mathbf{B} = B_0(1 + az)\mathbf{e}_z - \frac{aB_0\rho}{2}\mathbf{e}_\rho$$

in the cylindrical coordinates, here a is a constant. The loop remains horizontal while falling. The wire has the mass m and the resistance of the loop is R .

(a) Find the induced emf in the loop when the loop falls down with the velocity v .

(b) Assuming that $a > 0$, determine the direction of the induced current. Explain.

(c) Show that the terminal velocity of the loop is $v_t = \frac{mgR}{a^2 B_0^2 \pi^2 r^4}$.



5. (20pts). A CD (thin insulating disk with a hole in the middle, inner radius a , out radius b) spins counter-clockwise in the x-y plane with angular velocity ω . The mass of the CD disk is m . A total charge Q is distributed uniformly on the surface of the disk.

(a) Find the magnetic moment of the spinning disk.

(b) Find the angular momentum and the gyromagnetic ratio of the spinning disk.

(c) Suppose a uniform magnetic field B is applied in the y-direction. What is the angular velocity with which the disk will precess?

6. (20pts) A circular parallel-plate capacitor of radius R and plate separation d is connected to a source of potential difference $V = V_0 \cos \omega t$. The edge effects can be ignored since $d \ll R$.
- (a) Find the electric field inside the capacitor and the area densities of the electric charge on the plates as a function of time.
 - (b) Find the magnetic field inside the capacitor as a function of time and the distance to the axis of the plate.
 - (c) Find the electromagnetic field energy flow density (the Poynting vector) through the lateral surface of the capacitor.